

Esophagus and Stomach

Department of Anatomy

The abdomen (4)

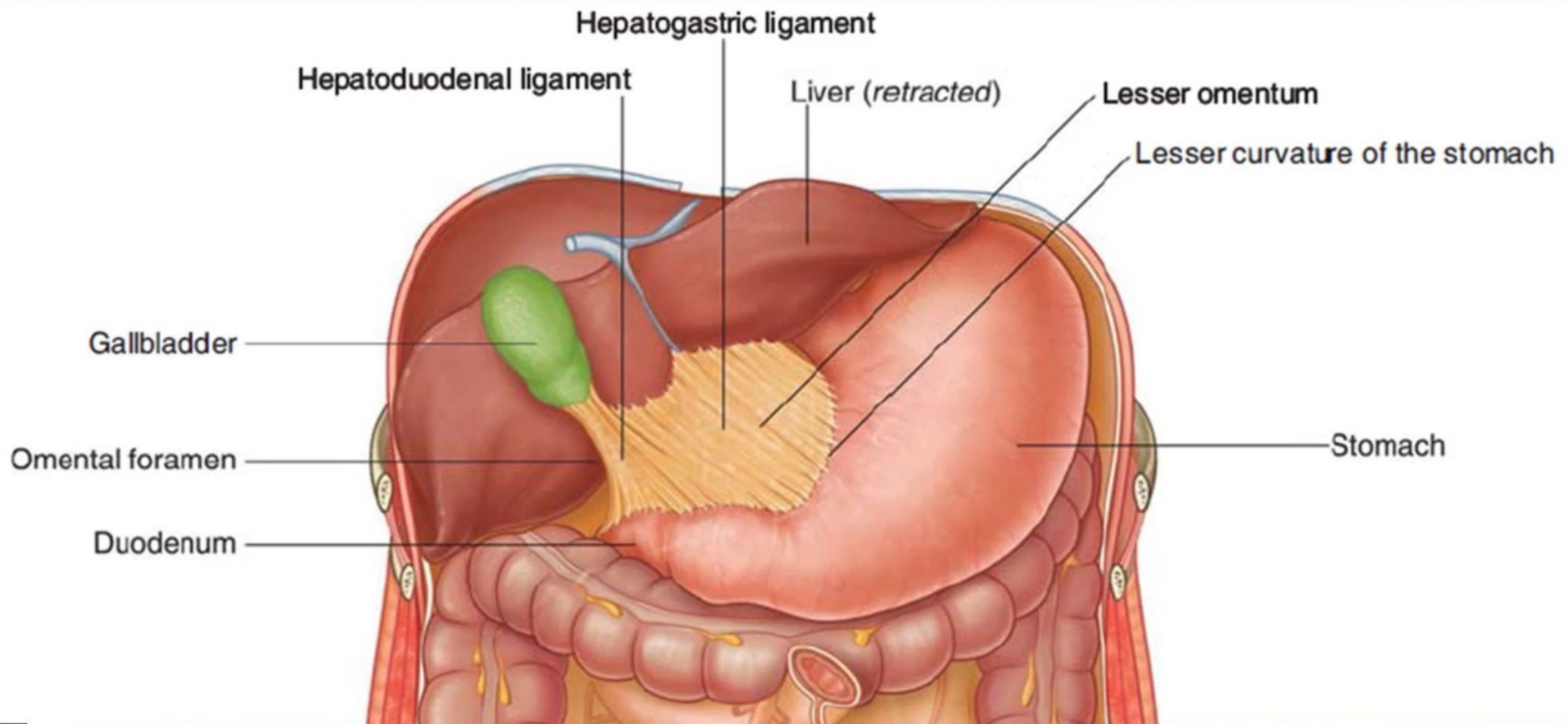
Second Part

Dr. Osman Ahmed Osman

Lecture Points

- **Overview of the Digestive System.**
 - **Gastrointestinal Functions.**
 - **Introduction of Esophagus (Location and course).**
 - **Relation of esophagus.**
 - **abdominal esophagus.**
 - **esophageal constrictions.**
 - **Blood supply of the esophagus.**
 - **Innervation of esophagus.**
 - **Lymphatic drainage of esophagus.**
 - **Function of esophagus and.**
- **Stomach**
 - **Function of the Stomach.**
 - **Location and Description.**
 - **Covering with peritoneum**
 - **Covering with peritoneum and lesser omentum.**
 - **Parts of the Stomach.**
 - **Cardia Opening.**
 - **Pylorus.**
 - **Muscular layers.**
 - **The stomach bed.**
 - **Nerve supply of the stomach.**
 - **Arterial supply stomach.**
 - **Venous drainage of the stomach.**
 - **Lymph Drainage of the stomach**

Stomach



Function of the Stomach

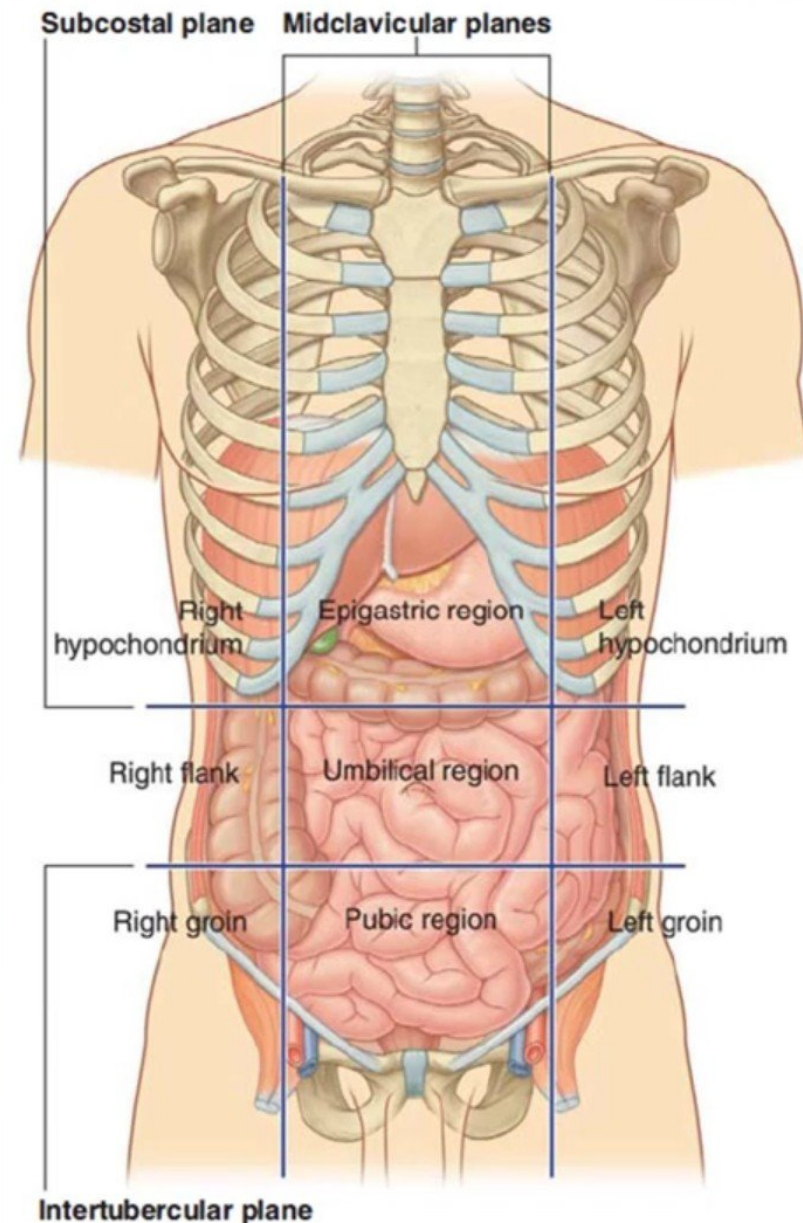
The stomach is the dilated portion of the alimentary canal and has three main functions:

- It stores food (in the adult it has a capacity of about 1500 mL).
- it mixes the food with gastric secretions to form a semifluid **chyme**.
- it controls the rate of delivery of the chyme to the small intestine so that efficient digestion and absorption can take place.

Stomach

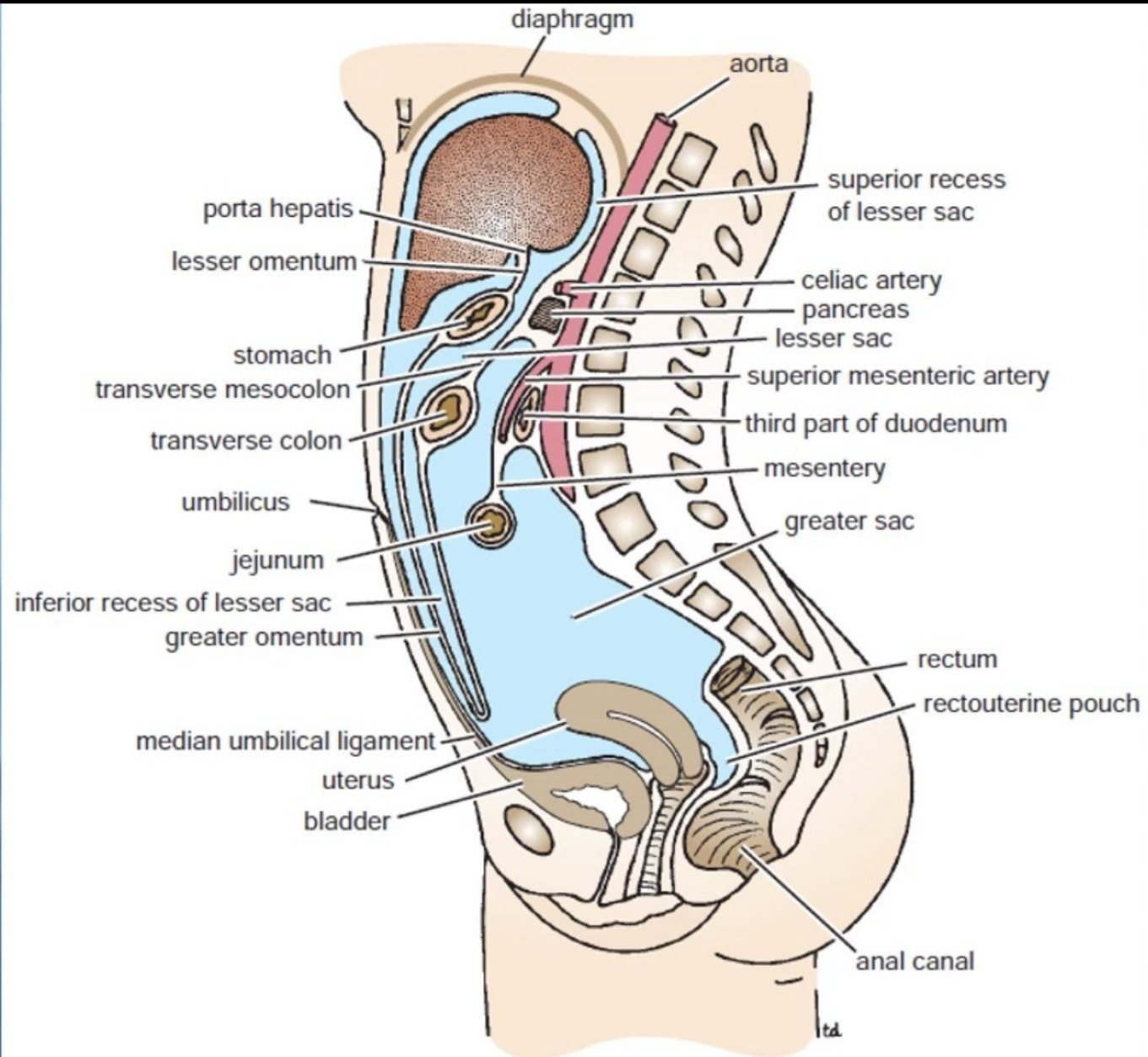
Location and Description

- It is **located** in the **left hypochondriac** and **epigastric regions** of the abdomen.
- It **extending** from beneath **the left costal margin** region into **the epigastric** and **umbilical regions**.
- Much of the stomach lies under **cover of the lower ribs**.
- It is **roughly J-shaped**.
- It has **two openings**, the **cardiac** and **pyloric orifices**.
- It has **two curvatures**, the **greater** and **lesser curvatures**.
- It has **two surfaces**, an **anterior** and a **posterior surface**.
- The stomach is **relatively fixed** at **both ends** but is **very mobile in between**.

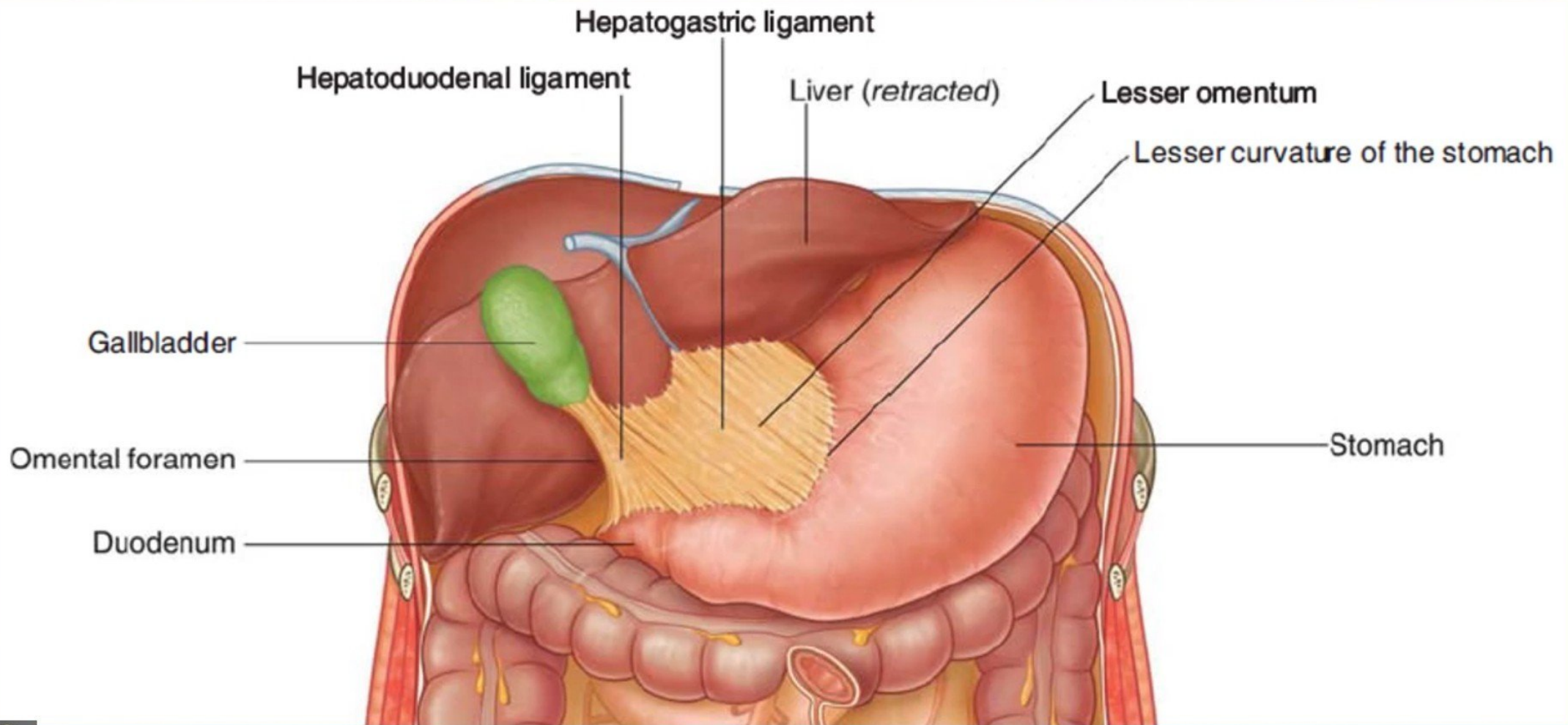


Covering with peritoneum

- The stomach is **relatively fixed at both ends** but is **very mobile in between**.
- The stomach is **completely invested by peritoneum**, which passes in a **double layer from the lesser curvature to the liver** as the **lesser omentum**, and hangs down from the **fundus** and **greater curvature** as the **greater omentum**.



Covering with peritoneum and lesser omentum



Parts of the Stomach

The stomach is divided into the following parts:

Cardia.

Fundus:

- It is dome-shaped.
- It is usually full of gas.

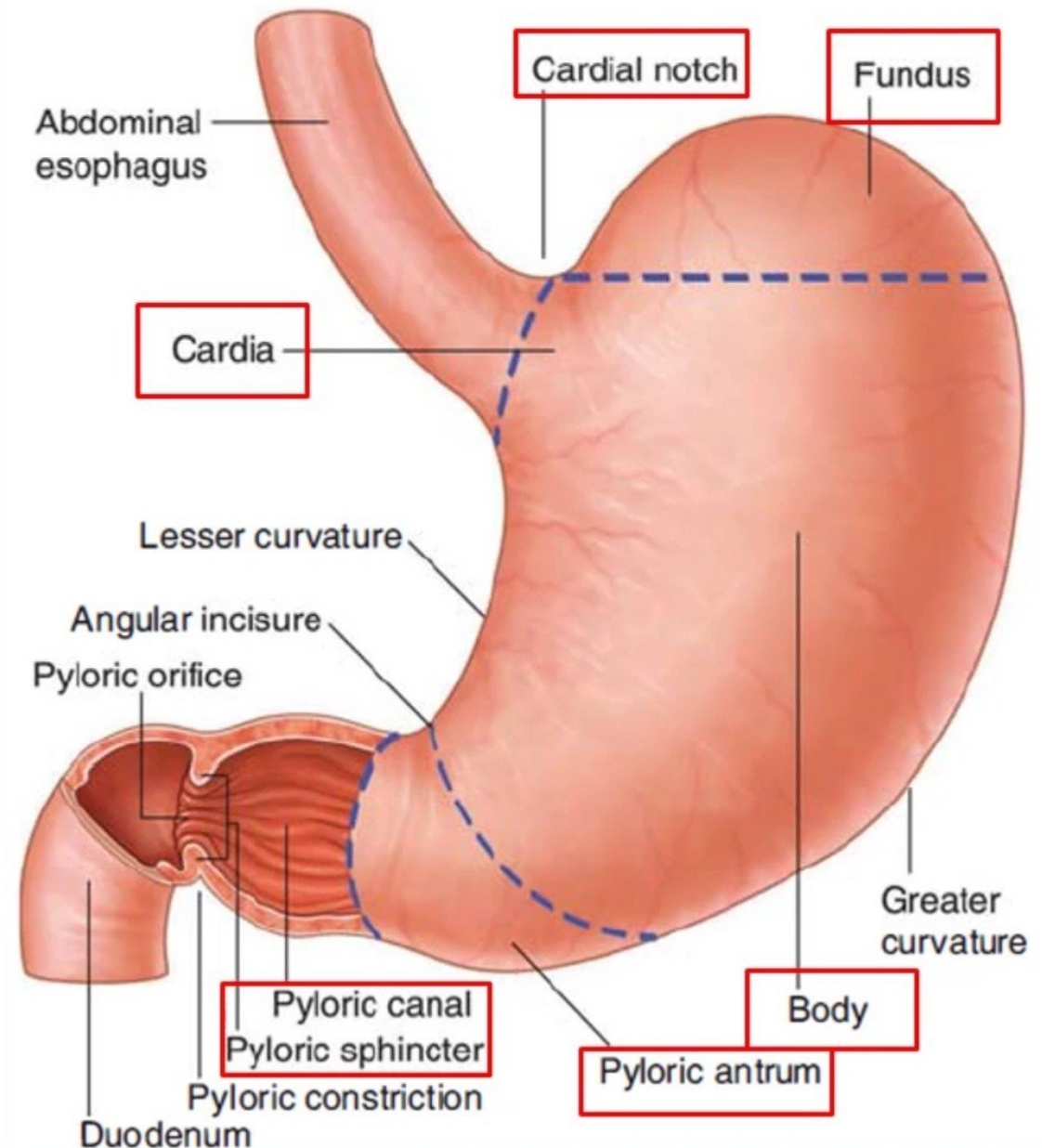
Body:

- This extends from the level of **the cardiac orifice** to the level of the **incisura angularis**, a constant notch in the lower part of the lesser curvature.

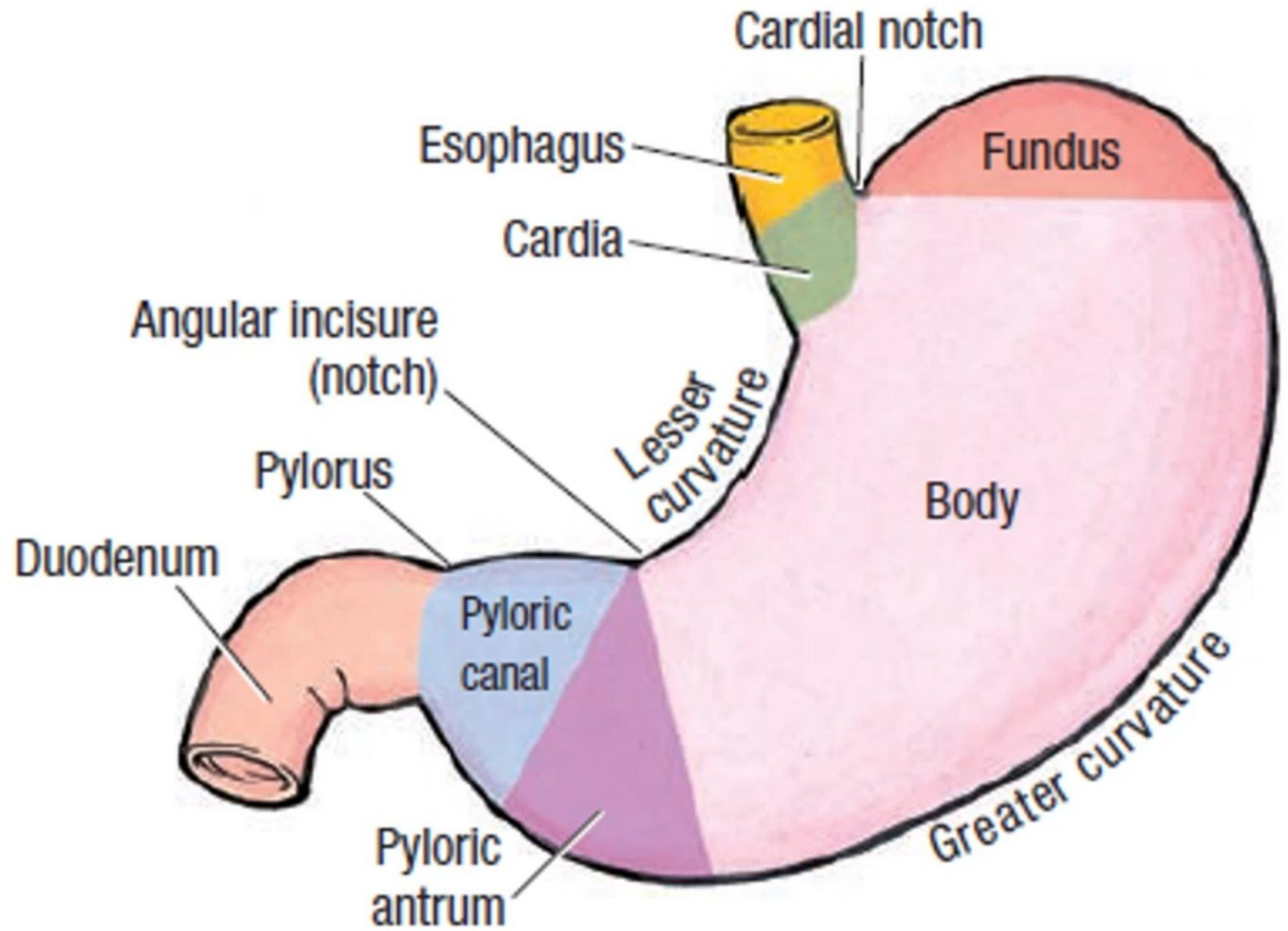
Pyloric antrum:

- This extends from the **incisura angularis** to the **pylorus**.

Pylorus.

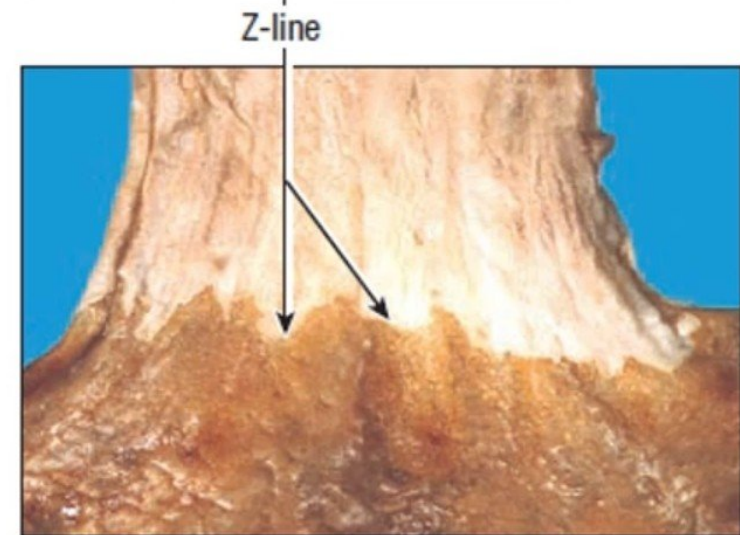
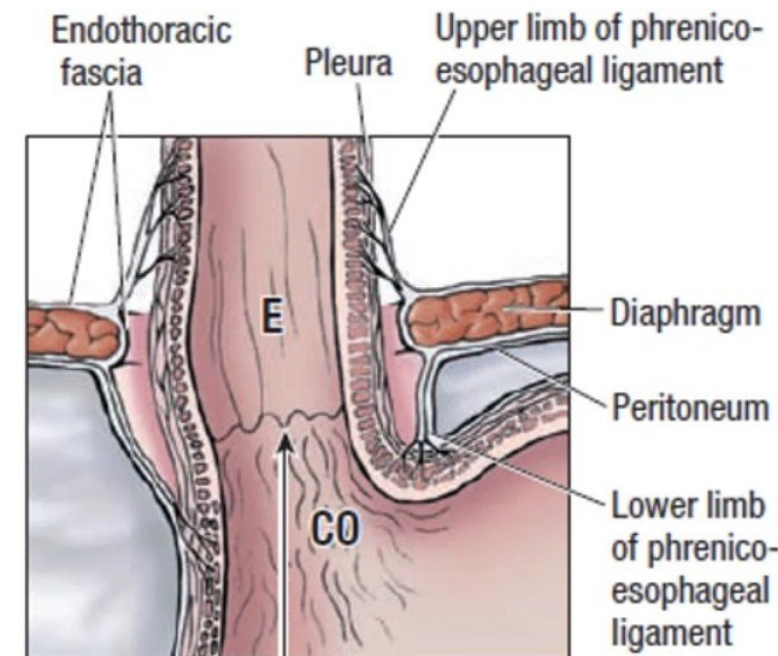


Parts of the Stomach



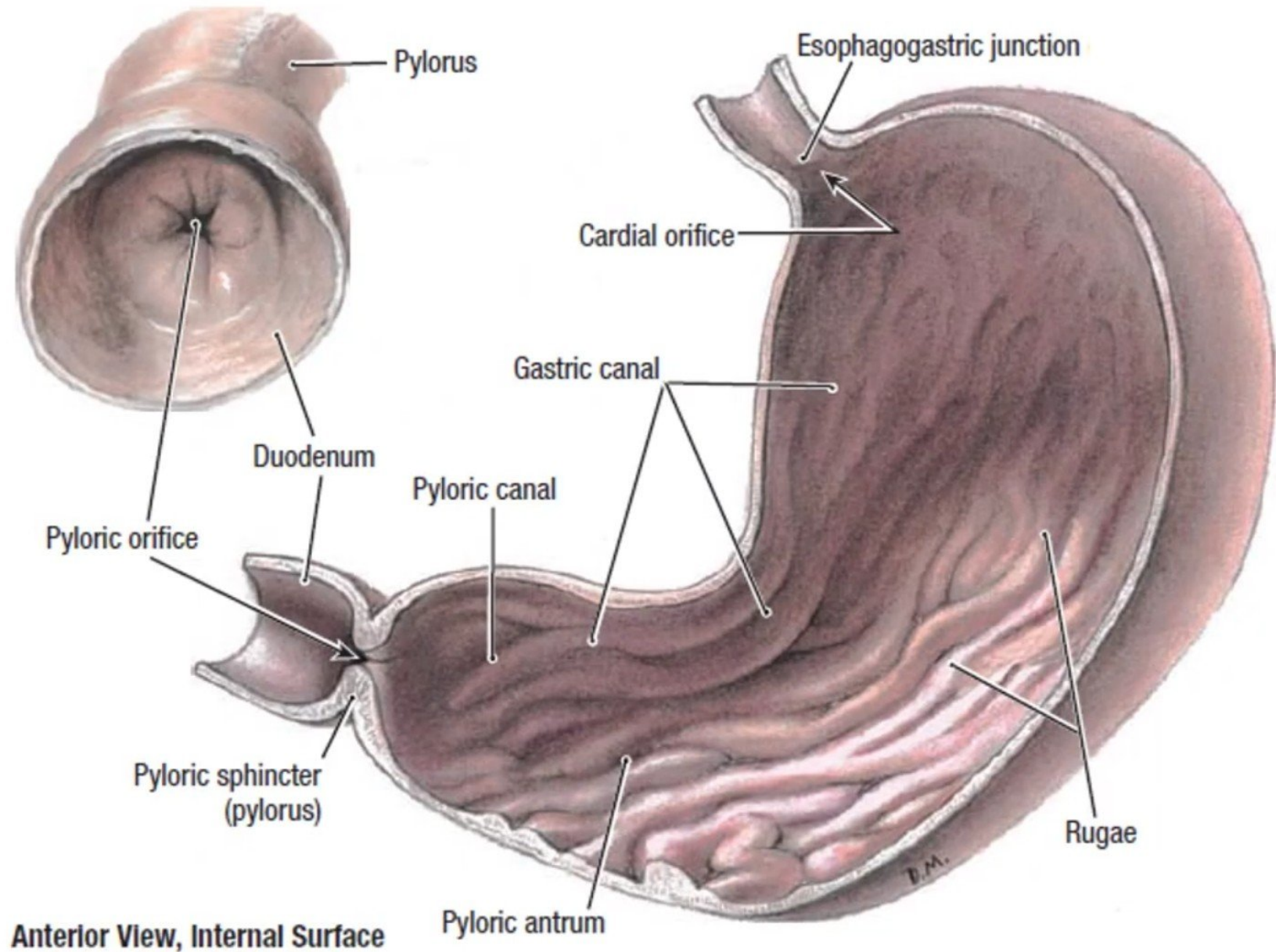
Cardia Opening

- It is **gastro-oesophageal junction**.
- It is most **fixed part** of the organ.
- It lies **2.5 cm to the left of the midline at the level of T10 vertebra**, behind the **seventh left costal cartilage**.
- It is **40 cm from the incisor teeth**.
- The **Z- line the stratified squamous epithelium** of the esophagus (**white** portion of photograph) to **simple columnar epithelium** of the stomach (**dark** portion).



Pylorus

- This is the most **tubular part** of the stomach.
- The **thick muscular wall** is called the **pyloric sphincter**, and the cavity of the pylorus is the **pyloric canal**.
- It is the **gastroduodenal junction**.
- It is position with the **stomach empty** is usually a **little to the right of the midline** at **the level of L1 vertebra** but may be lower.



Anterior View, Internal Surface

The lesser curvature and greater curvature

The lesser curvature

- It **forms** the right border of the stomach.
- It **extends** from **the cardiac orifice** to the **pylorus**.
- It is **suspended** from **the liver** by the **lesser omentum**.

The greater curvature:

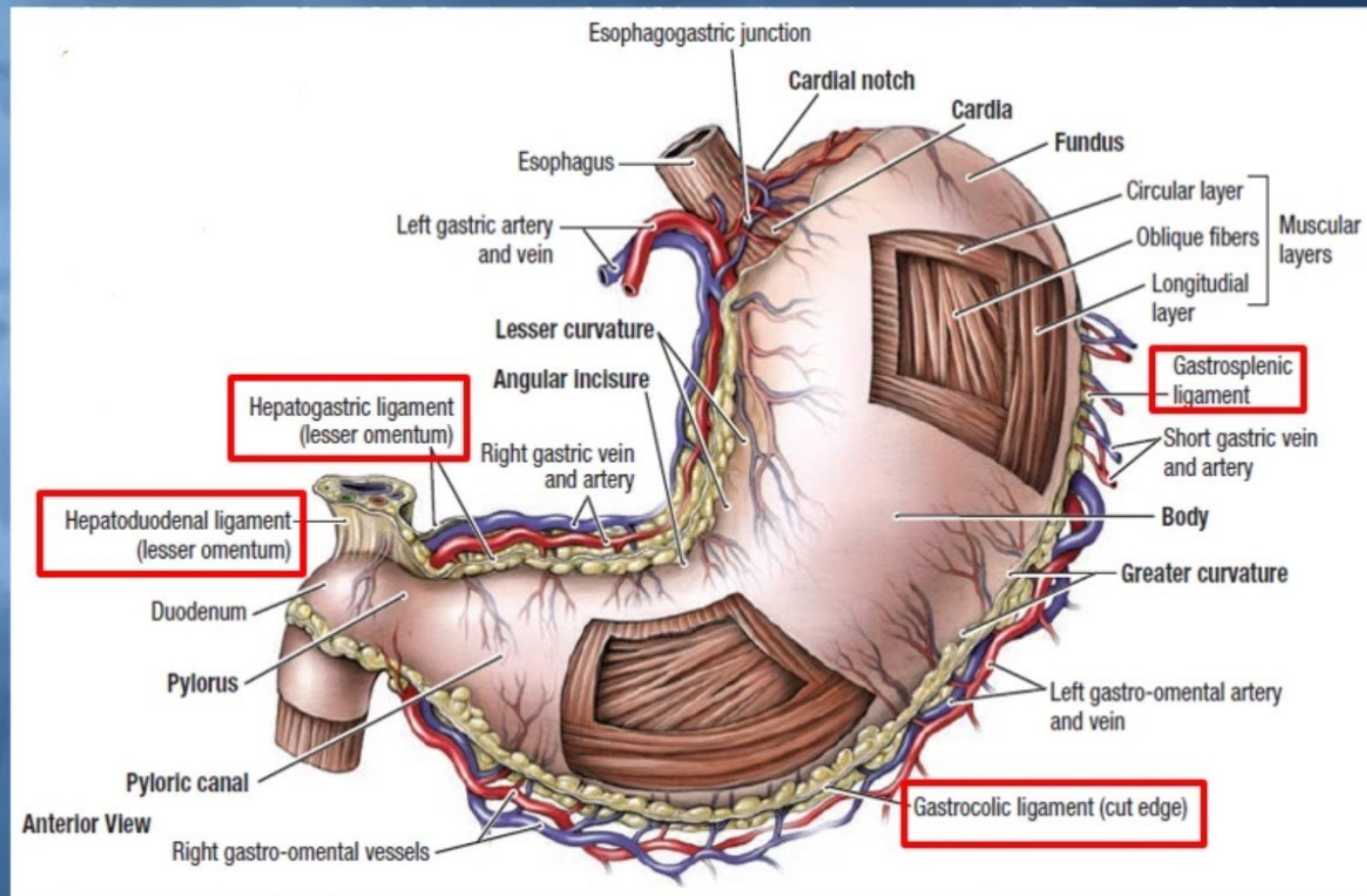
- It is much **longer** than **the lesser curvature**.
- It extends from **the left of the cardiac orifice**, over the **dome of the fundus**, and along the left border of the stomach **to the pylorus**.

The gastrosplenic omentum (ligament)

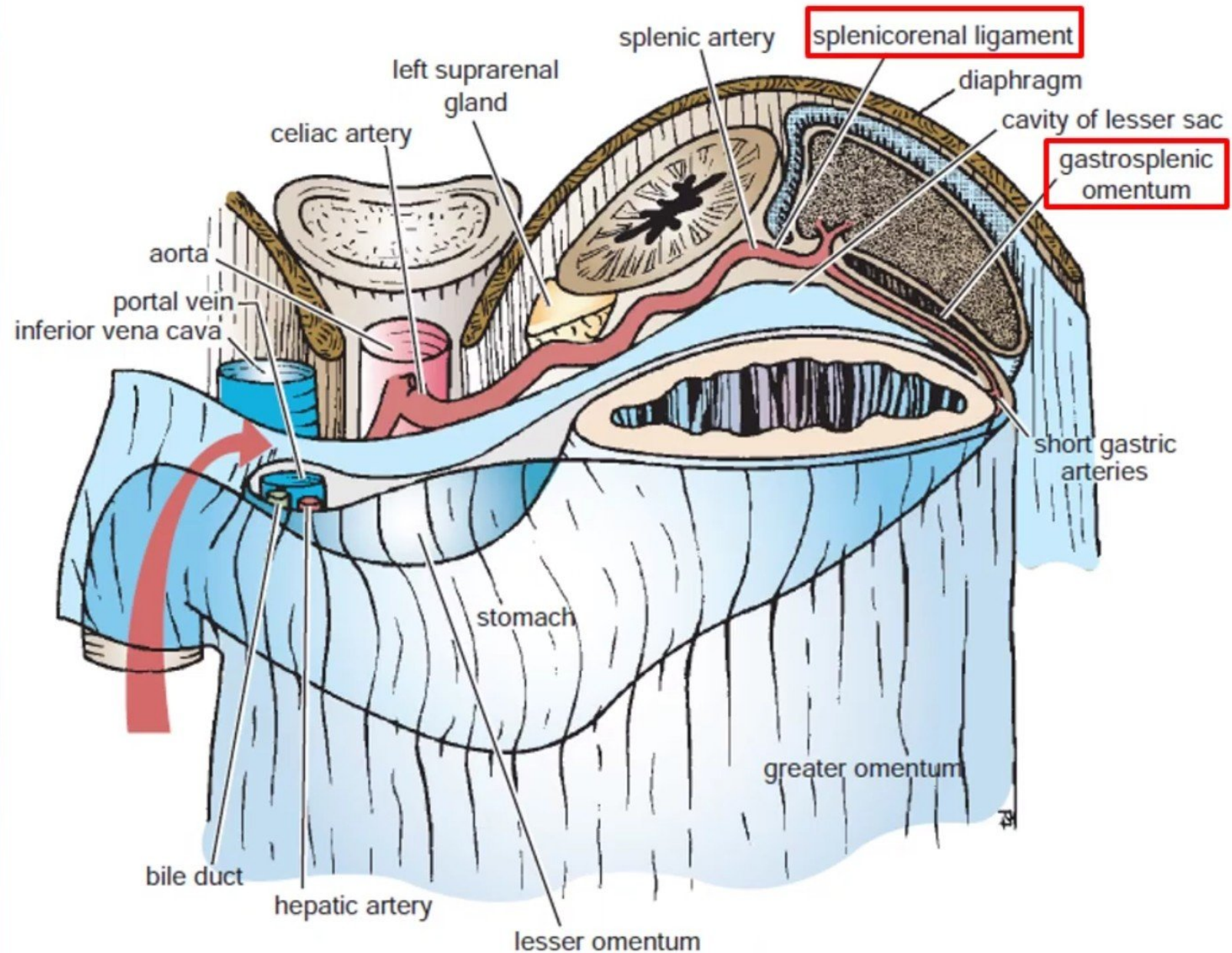
- It **extends** from **the upper part of the greater curvature** to **the spleen**, and

The greater omentum

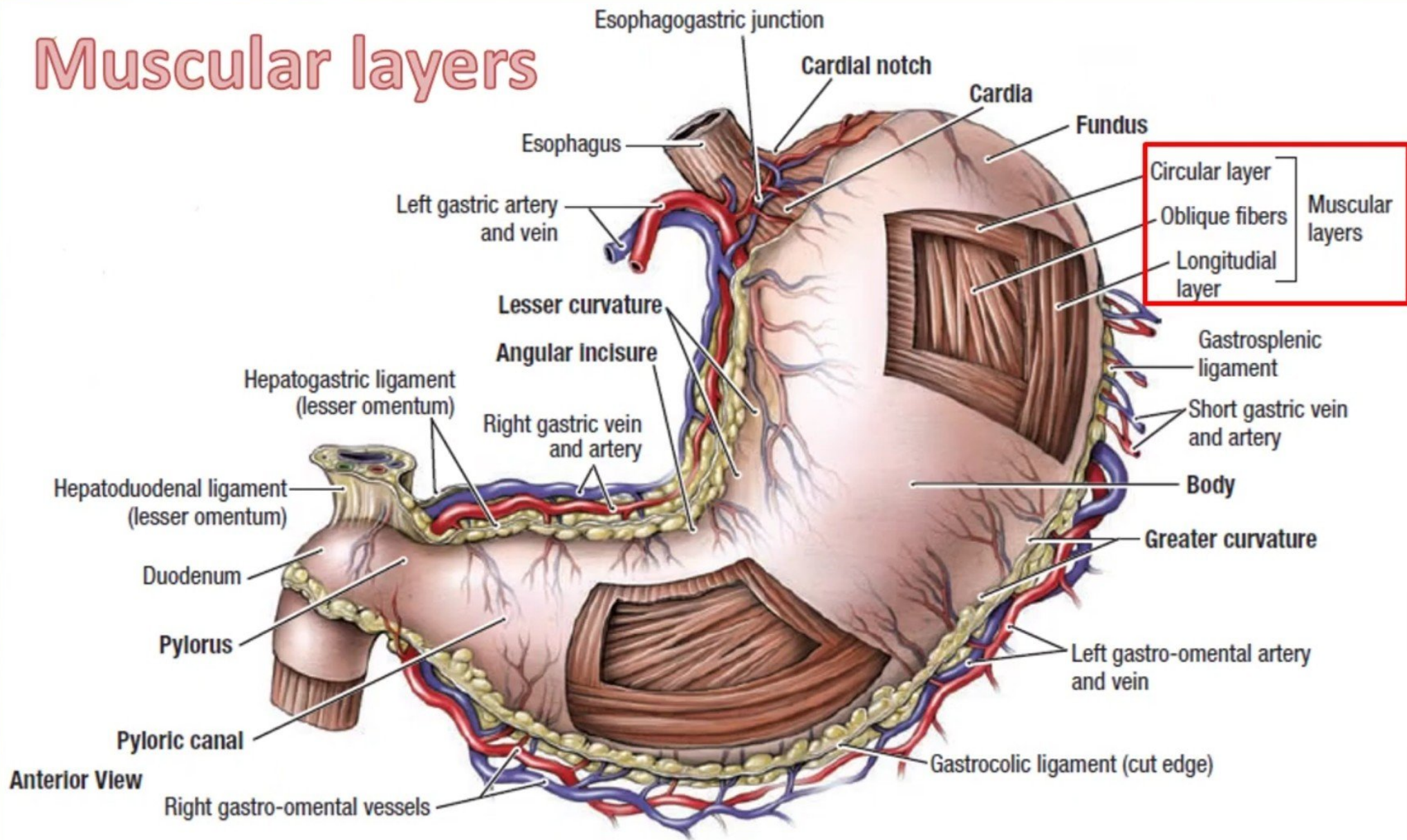
- It **extends** from **the lower part of the greater curvature** to the **transverse colon**.



Transverse section of the lesser sac showing the **arrangement of the peritoneum** in the formation of **the lesser omentum**, the gastrosplenic omentum, and **the splenicorenal ligament**. **Arrow** indicates the position of the **opening of the lesser sac**.



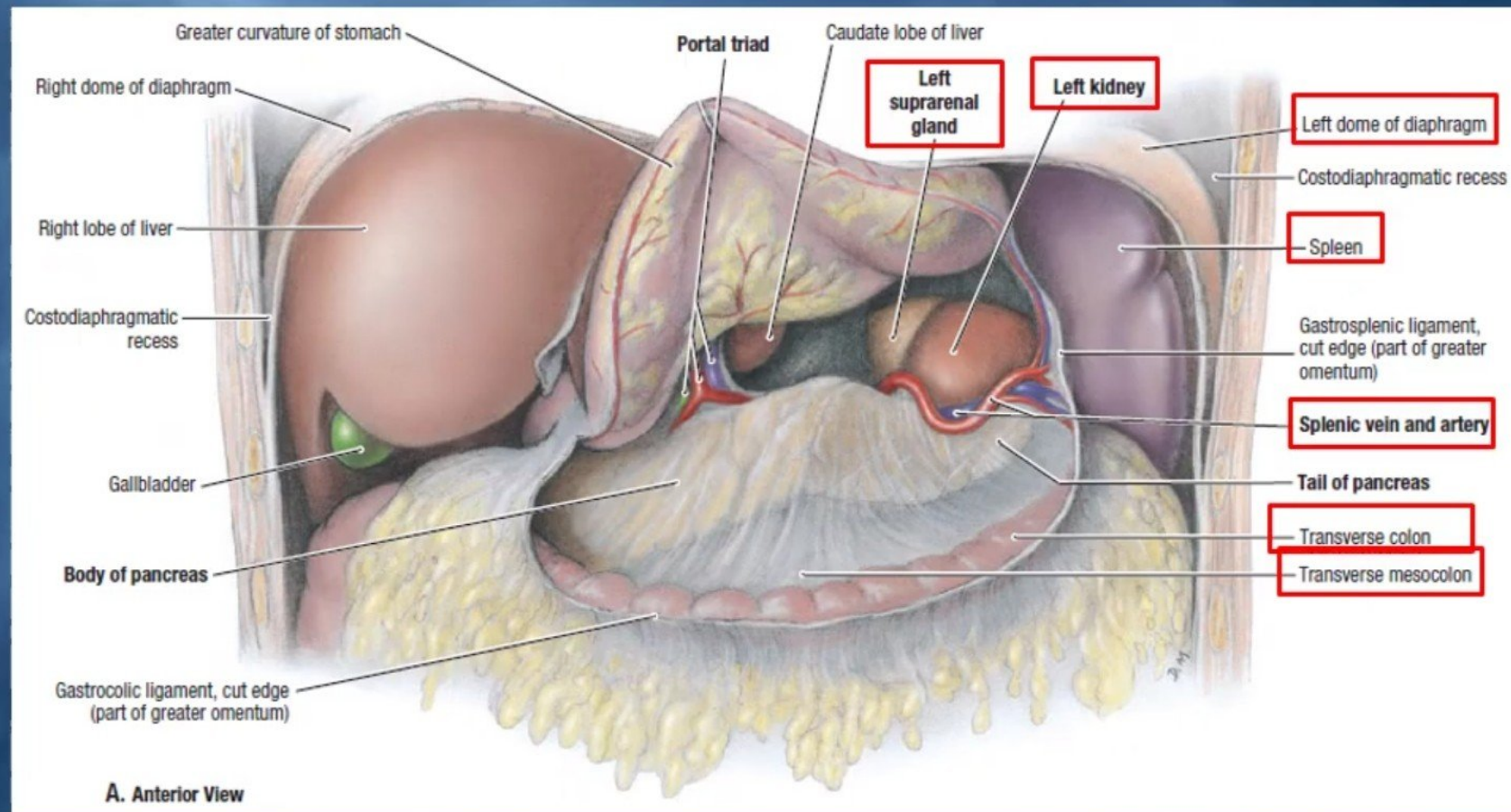
Muscular layers



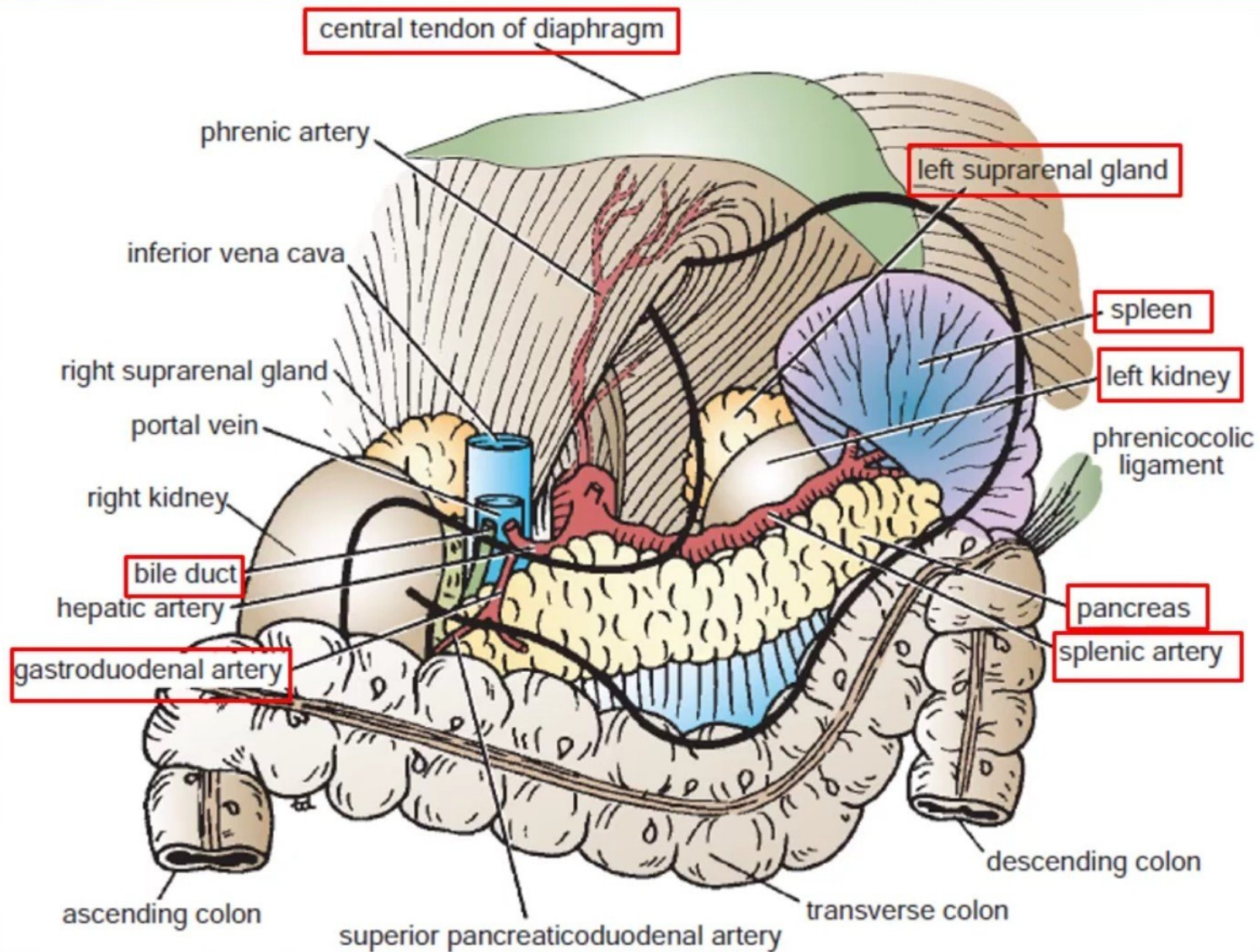
The stomach bed is formed by:

- ✓ Pancreas.
- ✓ Spleen.
- ✓ Left kidney.
- ✓ Left suprarenal gland,
- ✓ Transverse colon and its mesocolon.
- ✓ Diaphragm.

The stomach bed



The stomach bed



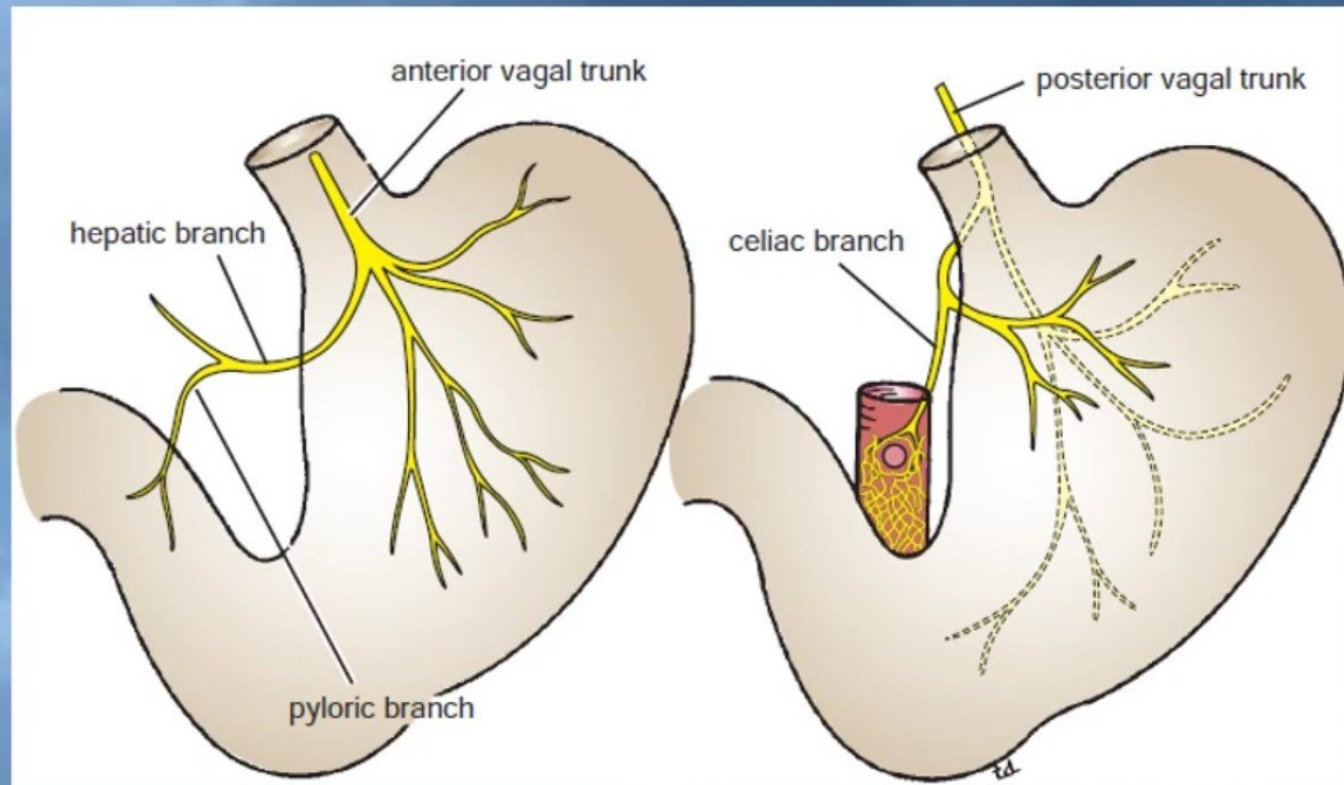
Nerve supply of the stomach

The anterior vagal trunk:

- It is formed in the thorax mainly from the **left vagus nerve**, enters the abdomen on the anterior surface of the esophagus.

The posterior vagal trunk:

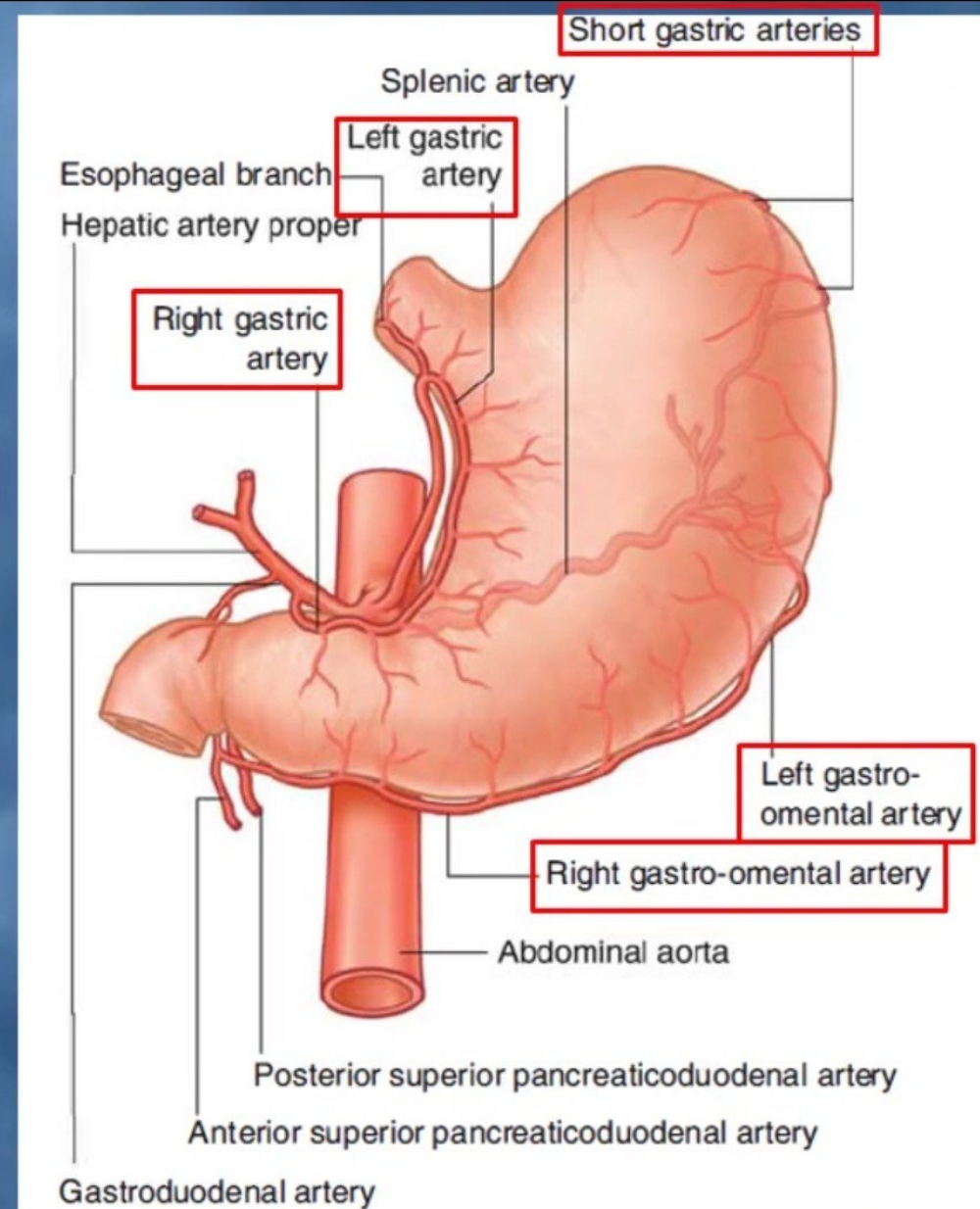
- It is formed in the thorax mainly from the **right vagus nerve**, enters the abdomen on the posterior surface of the esophagus.



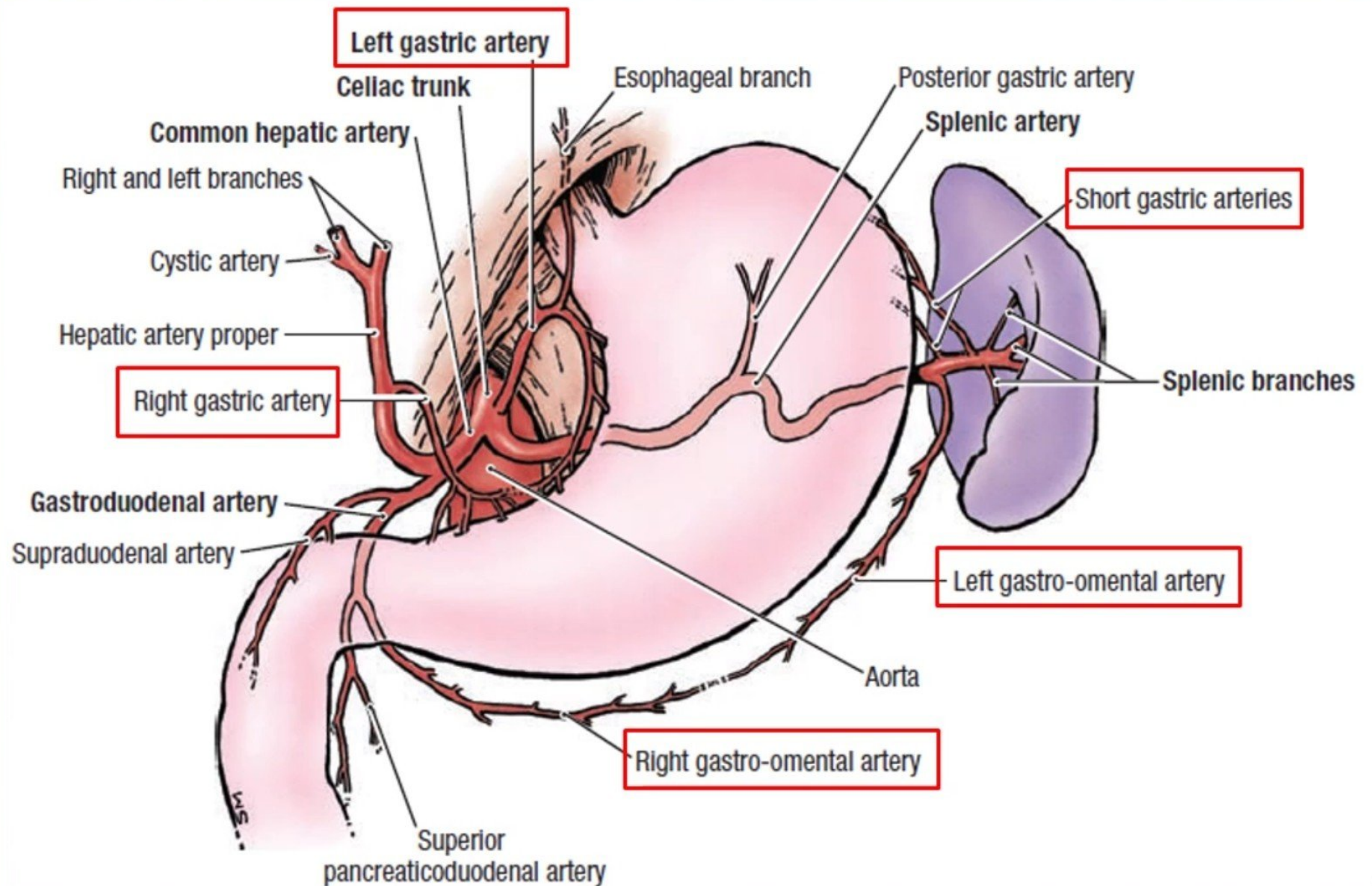
- The sphincter is **constricted** by **sympathetic stimulation** and **relaxed** by **parasympathetic** action.
- The **sympathetic innervation** of the stomach carries a proportion of **pain-transmitting nerve fibers**, whereas the **parasympathetic vagal fibers** are **secretomotor to the gastric glands** and **motor to the muscular wall of the stomach**.

Arterial supply stomach

- ✓ The **left gastric artery** arises from the celiac artery.
- ✓ The **right gastric artery** arises from the hepatic artery.
- ✓ The **short gastric arteries** arise from the splenic artery at the hilum of the spleen.
- ✓ The **left gastroepiploic artery** arises from the splenic artery at the hilum of the spleen.
- ✓ The **right gastroepiploic artery** arises from the gastroduodenal branch of the hepatic artery.

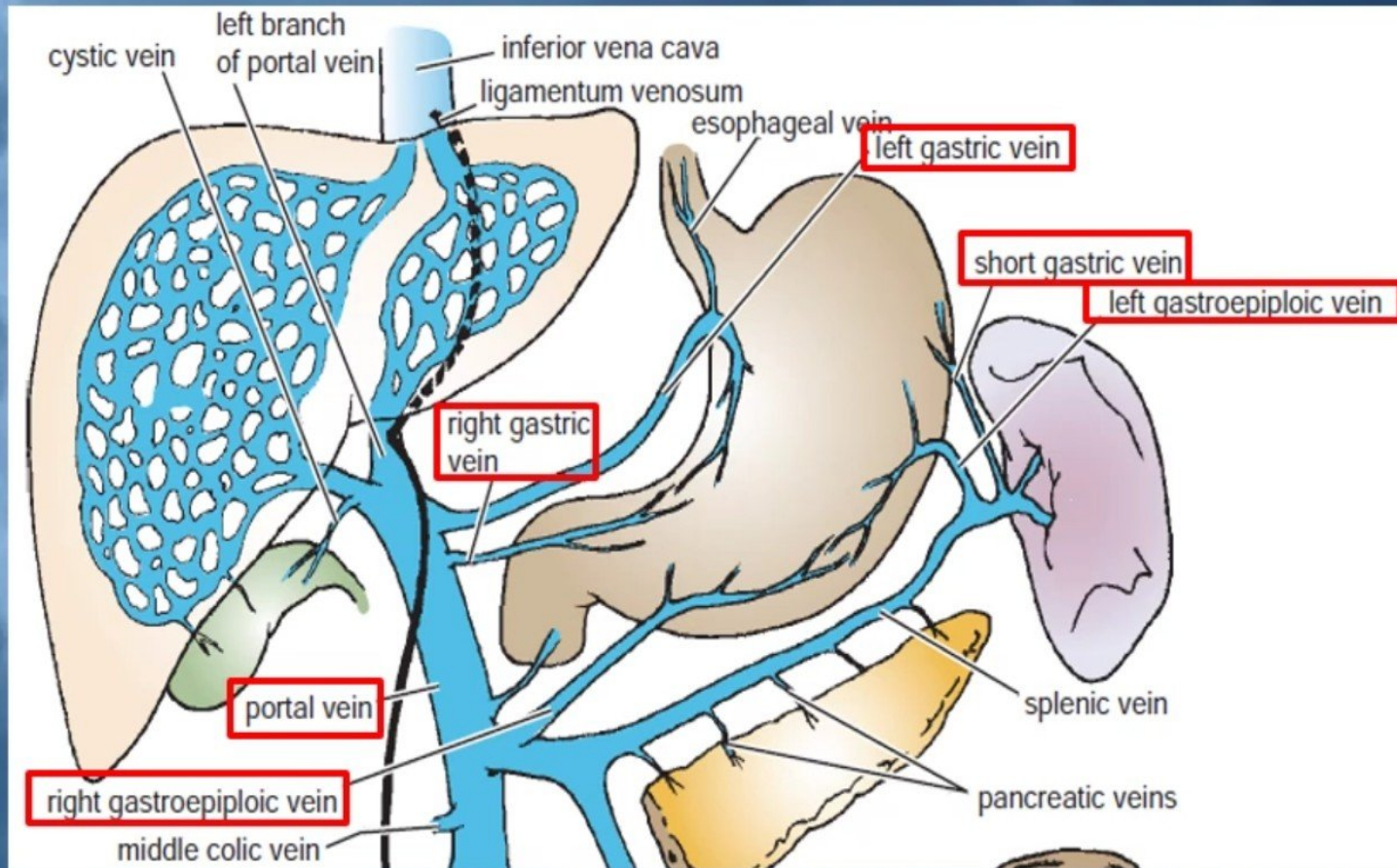


Arterial supply of stomach



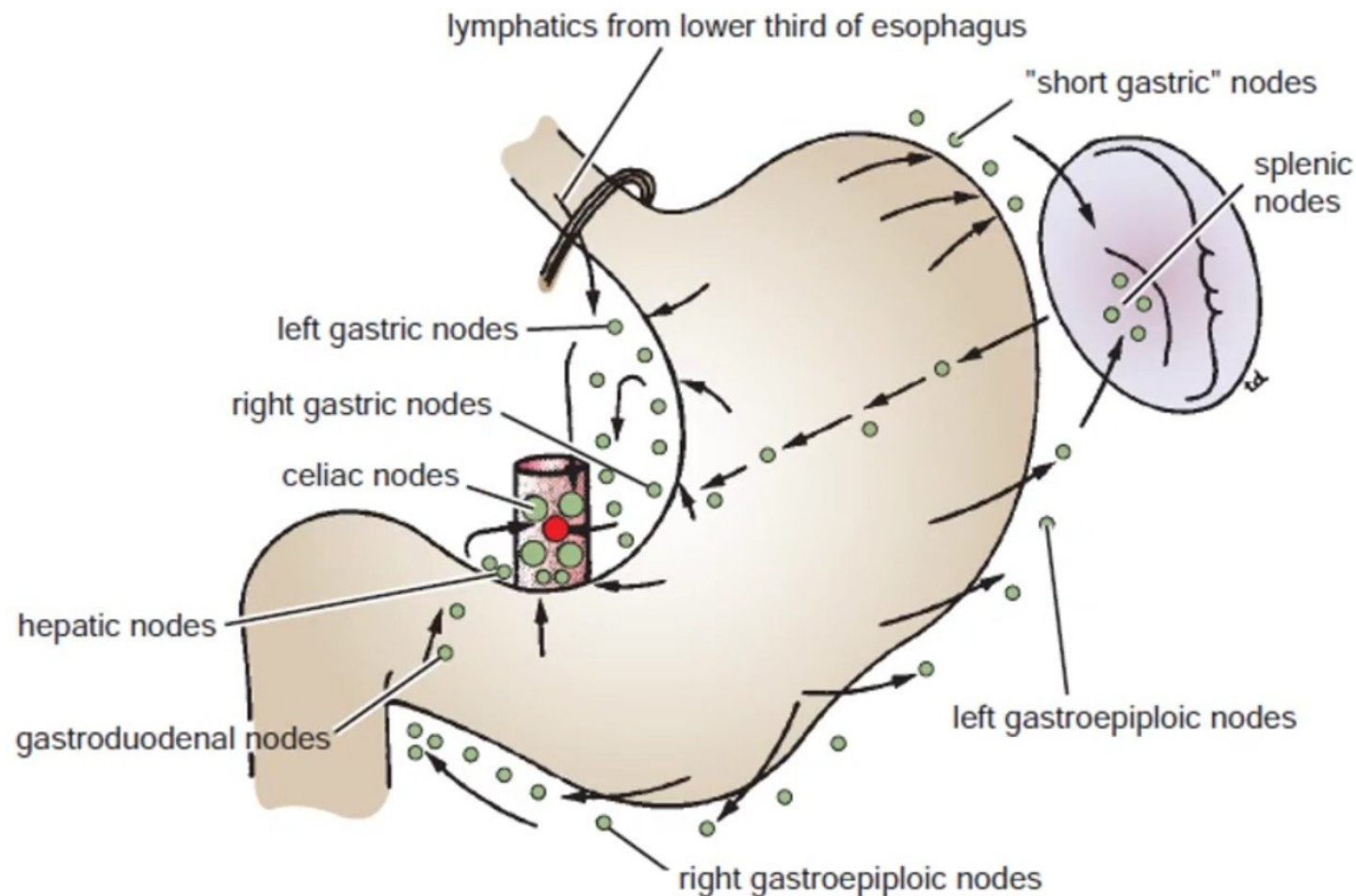
Venous drainage of the stomach

- The **veins drain into the portal circulation.**
- The **left** and **right gastric veins** drain directly into the portal vein.
- The **short gastric veins** and the **left gastroepiploic veins** join the splenic vein.
- The **right gastroepiploic vein** joins the superior mesenteric vein.



Lymph Drainage of the stomach

- The lymph vessels follow the arteries into the left and **right gastric nodes**, the **left and right gastroepiploic nodes**, and the **short gastric nodes**.
- All lymph from the stomach eventually passes to **the celiac nodes** located around the root of the celiac artery on the posterior abdominal wall.



Thanks